INTRODUCTION

1.1 INTRODUCTION:

Global import and export activities generate large volumes of trade data every day. Analyzing this data manually is complex and time-consuming.
This project uses data collected from the TradeStat government website and applies data analtytics techniques to analyse trade, product performance, and country-wise trade statistics.

In addition to this, the rapid growth of international trade has made it essential to manage and analyze large datasets efficiently. Countries across the world are continuously exchanging goods and services, which generates massive amounts of data related to imports, exports, pricing, demand, and supply. Handling such vast data manually becomes difficult and may lead to errors and delays.

With the advancement of technology, **data analytics** has become a powerful tool for extracting useful insights from raw data. It helps in identifying trends, patterns, and relationships within the data, which are not easily visible through manual analysis. By using analytical tools and techniques, businesses and governments can make better decisions regarding trade policies, market strategies, and economic planning.

Furthermore, understanding trade patterns is very important for economic growth. By analyzing import and export data, it is possible to identify which products are in high demand, which countries are key trading partners, and how trade trends change over time. This information can be useful for businesses to expand their markets and improve their performance.

The *Import Export Trade Analytics System* is designed to overcome the limitations of manual data analysis by providing an automated and efficient solution. The system collects trade data, processes it, and presents the results in a clear and understandable format. It uses data visualization techniques such as charts and graphs to make the analysis more effective

1.2 OBJECTIVE:

1. To collect import–export product data from TradeStat:

The first objective of this project is to collect import and export data from reliable government sources such as TradeStat. This ensures that the data used in the system is accurate, authentic, and well-structured. Collecting data from trusted sources helps in maintaining data quality and reliability for further analysis.

2. To analyze trade data using data analytics techniques:

Another important objective is to analyze the collected data using various data analytics techniques. The system processes raw data by cleaning, organizing, and filtering it to extract meaningful information. This helps in identifying patterns, trends, and relationships within the trade data.

3. To identify top importing and exporting countries:

This project aims to identify the leading countries involved in import and export activities. By analyzing country-wise trade data, the system can determine which countries have the highest trade volumes and understand their role in global trade.

4. To visualize trade trends using graphs and charts:

The system also focuses on presenting data in a visual format using graphs and charts. Visualization makes it easier to understand complex data and helps users quickly identify trends, comparisons, and changes over time.

5. To provide meaningful insights for decision-making:

The final objective is to generate useful insights from the analyzed data. These insights can help businesses, researchers, and policymakers make better decisions related to trade strategies, market expansion, and economic planning.

1.3 STATEMENT OF PROBLEM:

Manual collection and analysis of import–export data is inefficient and prone to errors. There is no automated system to fetch, process, and analyze large trade datasets effectively. This creates difficulty in understanding trade trends and making accurate decisions.

1. Inefficiency of manual data collection:

In the existing system, import and export data is collected manually from different sources. This process requires a lot of time and effort. As the volume of trade data increases daily, manual collection becomes difficult and slow.

2. High chances of errors:

Since the data is handled manually, there is a higher chance of mistakes during data entry and processing. Even small errors can affect the final analysis and lead to incorrect results.

3. Lack of automation:

The absence of automation makes the system less efficient. Tasks like data cleaning, sorting, and analysis must be done manually, which reduces productivity and increases workload.

4. Difficulty in analyzing large datasets:

Trade data is large and complex. Without proper tools, it becomes difficult to analyze and extract useful information. This limits the ability to identify trends and patterns.

REVIEW OF LITERATURE SURVEY

2.1 EXISTING SYSTEM & DISADVANTAGES OF EXISTING SYSTEM:

Existing System:

Data is collected manually from websites. Analysis is done using spreadsheets. There is limited data visualization and no automation or real-time data access.

In the existing system, import and export data is gathered manually from different online sources. The collected data is then analyzed using spreadsheet tools such as Microsoft Excel. Although this method is simple, it is not efficient for handling large volumes of data. The system lacks advanced analytical capabilities and does not support automation, making the overall process slow and less reliable.

The existing system has several disadvantages that affect its performance and usefulness.

1. Lack of proper data preprocessing and integration:

The existing system lacks proper data preprocessing and integration, resulting in inconsistent, incomplete, and low-quality trade data.

Since the data is collected manually, it may contain errors, missing values, and inconsistencies. Without proper cleaning and integration, the data cannot be used effectively for analysis, which reduces the accuracy and reliability of results.

2. No support for exploratory data analysis (EDA):

There is no support for exploratory data analysis (EDA), making it difficult to identify trade patterns and trends.

EDA is important for understanding the structure of data and discovering hidden patterns. Without it, users cannot easily analyze relationships, trends, or anomalies in trade data.

3. Inability to handle large volumes of data efficiently:

The system does not scale efficiently to handle large volumes of data. As import and export data increases, the system becomes slow and difficult to manage. This limits its ability to process and analyze large datasets effectively.

4. Limited analytical and visualization tools:

Limited analytical and visualization tools restrict meaningful interpretation. The system provides only basic tools, which are not sufficient to represent complex data clearly. This makes it difficult for users to understand the data and draw useful conclusions.

5. Absence of updated data leading to inaccurate decision-making:

Absence of updated data leads to inaccurate decision-making. Since the data is not updated regularly, users may rely on outdated information. This can result in incorrect analysis and poor decisions, especially in a fast-changing trade environment.

2.2 PROPOSED SYSTEM:

The proposed system collects import–export product data from the TradeStat government website. The collected data is cleaned, processed, and analyzed using data analytics tools. Graphical visualization techniques are used to represent trends clearly and effectively.

In this system, data is collected from reliable sources to ensure accuracy. The collected data is then cleaned and organized to remove errors and inconsistencies. This helps in improving the quality of the data for further analysis.

The system analyzes the processed data to identify patterns and trends in trade activities. It helps in understanding country-wise trade performance and product demand.

COMPANY OVERVIEW

3.1 Introduction of the company

DharwadHubballiTutor, established on **15 November 2020**, is a professionally driven EdTech and digital solutions organization focused on developing industry-ready talent and delivering high-impact technology services. Since its inception, the organization has been committed to bridging the gap between education and industry through practical, outcome-oriented learning and technology enablement.

Operating at the intersection of **education, data, and digital transformation**, DharwadHubballiTutor supports individuals and organizations in adapting, scaling, and succeeding in a rapidly evolving digital economy.

Our offerings include **professional training programs, corporate workshops, seminars, online learning solutions**, along with **software development, BI dashboard implementation, and digital marketing services**. Each engagement is designed to deliver **measurable outcomes, operational efficiency, and long-term value**.

3.2 Vision, Mission, Objectives, and Quality Policy Vision:

To provide advanced digital solutions for educational institutions through innovative software technologies that enhance learning outcomes and operational efficiency. Mission: To simplify educational management systems and improve academic efficiency using modern technologies while maintaining the highest standards of security and reliability.

Objectives:

- To deliver high-quality software solutions that meet client requirements.
- To automate educational processes and reduce manual workload.
- To improve student performance monitoring and analysis.
- To maintain data security and privacy standards.

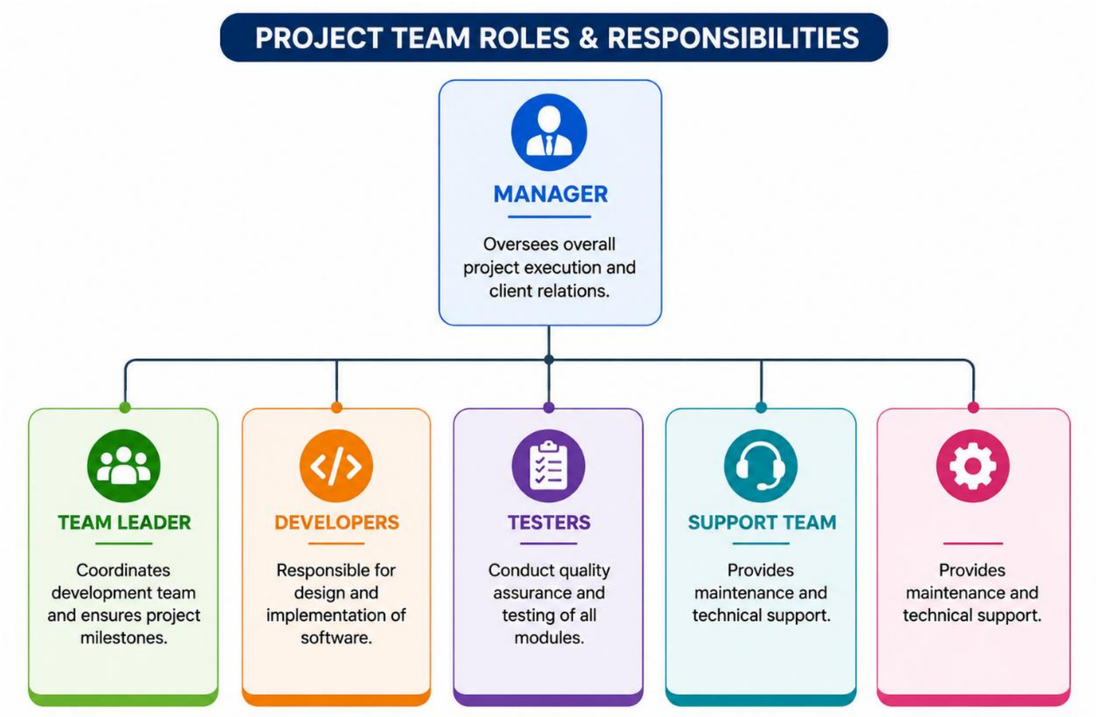
Quality Policy:

The company ensures software quality through proper testing, security implementation, reliability verification, and user-friendly design. All deliverables undergo rigorous quality assurance processes before deployment.

3.3 Company Profile and Organizational Structure Company Profile:

- DharwadHubballiTutor Software Development Company specializing in educational solutions.
- Specialization in Web Applications and custom development.
- Focus on Educational Software Development and management systems. Expertise in Database Management Systems design and implementation.

Organizational Structure:



3.4 Products and Services

Services Provided:

- **Web Application Development with modern frameworks and technologies.**
- **Database Design and optimization for performance.**
- **Educational Software Development tailored to institutional needs.**
- **Technical Support and maintenance of deployed systems.**
- **Training programs for end-users and administrators.**

3.5 SWOT Analysis

SYSTEM REQUIREMENT

4.1 SYSTEM REQUIREMENTS:

The proposed system requires both hardware and software components to perform data collection, processing, analysis, and visualization efficiently.

4.1 Hardware Requirements:

Processor: Intel i3 or above

RAM: 8 GB or above

Hard Disk: 256 GB or above

Internet Connection

The system requires a computer with at least an Intel i3 processor to handle data processing tasks. A minimum of 8 GB RAM is needed for smooth execution of data analysis tools. A 256 GB hard disk is sufficient for storing datasets and project files. An internet connection is essential to collect data from online sources like TradeStat..

4.2 Software Requirements:

Operating System: Windows

Python 3.10.12

VS Code

MySQL

Microsoft Power BI

The system uses Windows as the operating system for compatibility and ease of use. Python is used as the main programming language for data analysis and processing. VS Code is used as the development environment for writing and executing code.

MySQL is used to store and manage the collected trade data efficiently. In addition, Microsoft Power BI is used for data visualization. It helps in creating interactive dashboards, charts, and reports, making it easier to understand complex trade data and identify trends.

4.3 Functional Requirements:

The system should collect import–export data from TradeStat.

The system should clean and preprocess the data.

The system should analyze trade data to identify trends.

The system should generate graphs and charts.

The system should store data in a database.

4.4 Non-Functional Requirements:

The system should be fast and efficient.

The system should be user-friendly.

The system should handle large data effectively.

The system should provide accurate results.

The system should be reliable and easy to use.

DEVELOPMENT ENVIRONMENT

5.1 LANGUAGES AND TOOLS USED:

Python:

Used as the main programming language for data analysis and processing.

Pandas & NumPy:

Used for data cleaning, manipulation, and handling large datasets.

Matplotlib & Seaborn:

Used for creating graphs and charts to visualize trade data.

MySQL:

Used as the database to store and manage trade data efficiently.

Microsoft Excel:

Used for basic data handling and preprocessing.

Microsoft Power BI**:**

Used for advanced data visualization and creating interactive dashboards.

5.2 USER INTERFACE (UI/UX)

HTML5: Used to design the structure of web pages.

CSS3: Used to improve the visual appearance of the interface.

JavaScript (Vanilla): Used to add interactivity and dynamic features.

Python Flask: Used as the backend framework to connect frontend and database.

The interface is **simple, user-friendly, and easy to navigate**.

Data is displayed using **charts, graphs, and dashboards**.

5.3 SERVER

The system is developed and executed on a **local machine**.

No dedicated server is required.

The local system is sufficient for data analysis tasks.

Can be deployed on a **web server in future** for better accessibility.

5.4 DATABASE

MySQL is used as the database management system.

Stores data in a **structured format**.

Allows efficient **data storage and retrieval**.

Helps in handling **large datasets effectively**.

SYSTEM DESIGN

6.1 DATA FLOW DIAGRAM:

A Data Flow Diagram (DFD) is a graphical representation that shows how data moves through a system. It explains the flow of information between input, processing, storage, and output components. DFD helps in understanding the working of the system clearly and systematically.

In the *Import Export Trade Analytics System*, the DFD shows how import–export trade data is collected, processed, analyzed, stored, and displayed through dashboards and visual reports. The system accepts trade data from sources such as the TradeStat website and processes it using data analytics techniques.

The system analyzes trade data to identify country-wise trade statistics, product performance, import–export trends, and other useful insights. The processed information is then displayed using charts, graphs, and dashboards for better understanding and decision-making.

SYMBOLS:

External Entities:

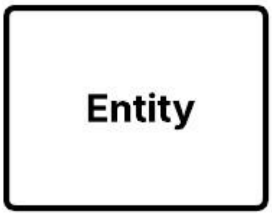
External entities are the sources or destinations of data outside the system. In this project, external entities include:

Users

Administrators

TradeStat Website

Data Analyst



Processes:

Processes transform input data into useful output. Major processes in this system include:

Data Collection

Data Cleaning and Preprocessing

Trade Data Analysis

Data Visualization

Report Generation

Data Stores:

Data stores are used to save important system data such as:

Import–Export Trade Data

User Information

Analytics Reports

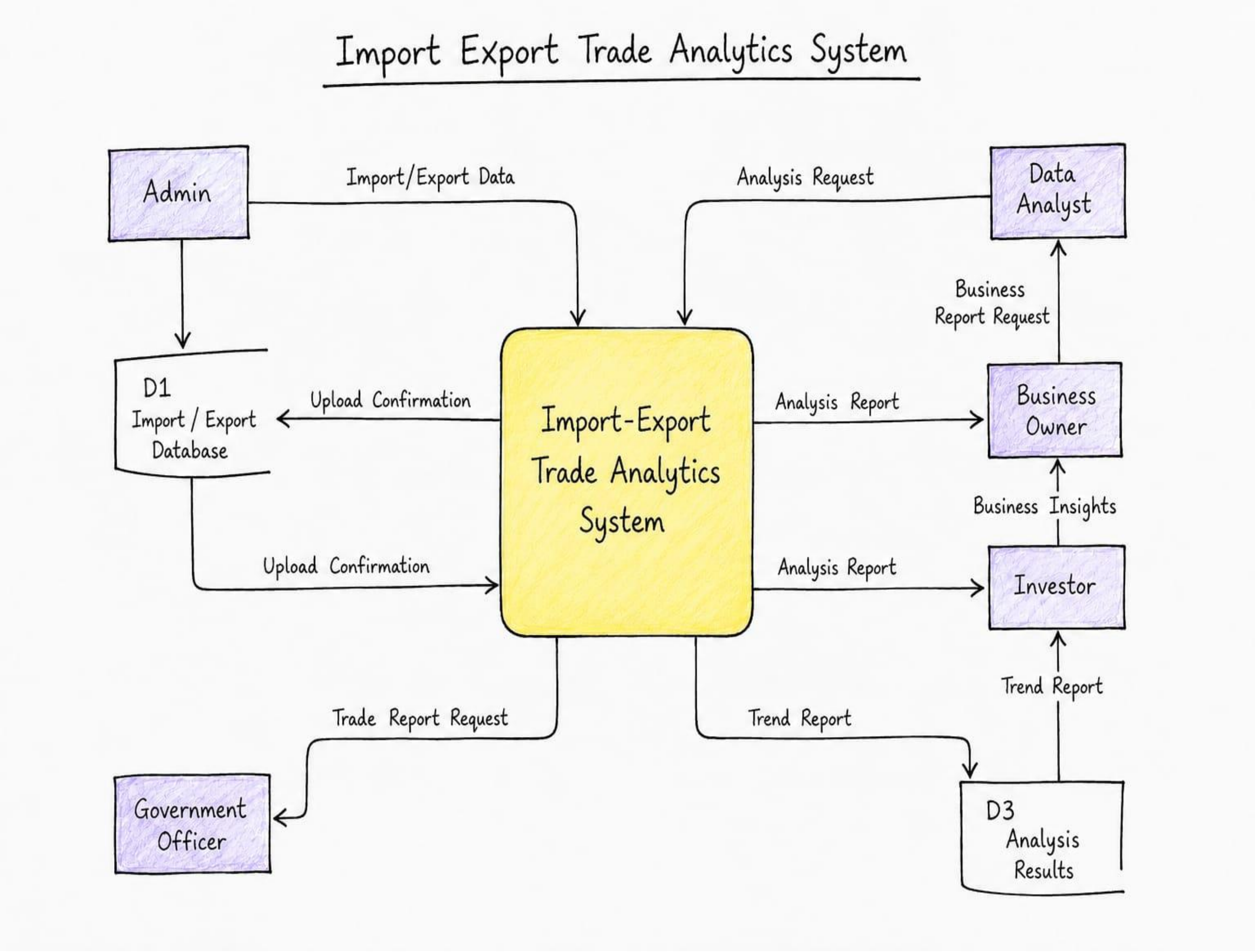
Dashboard Data

Or

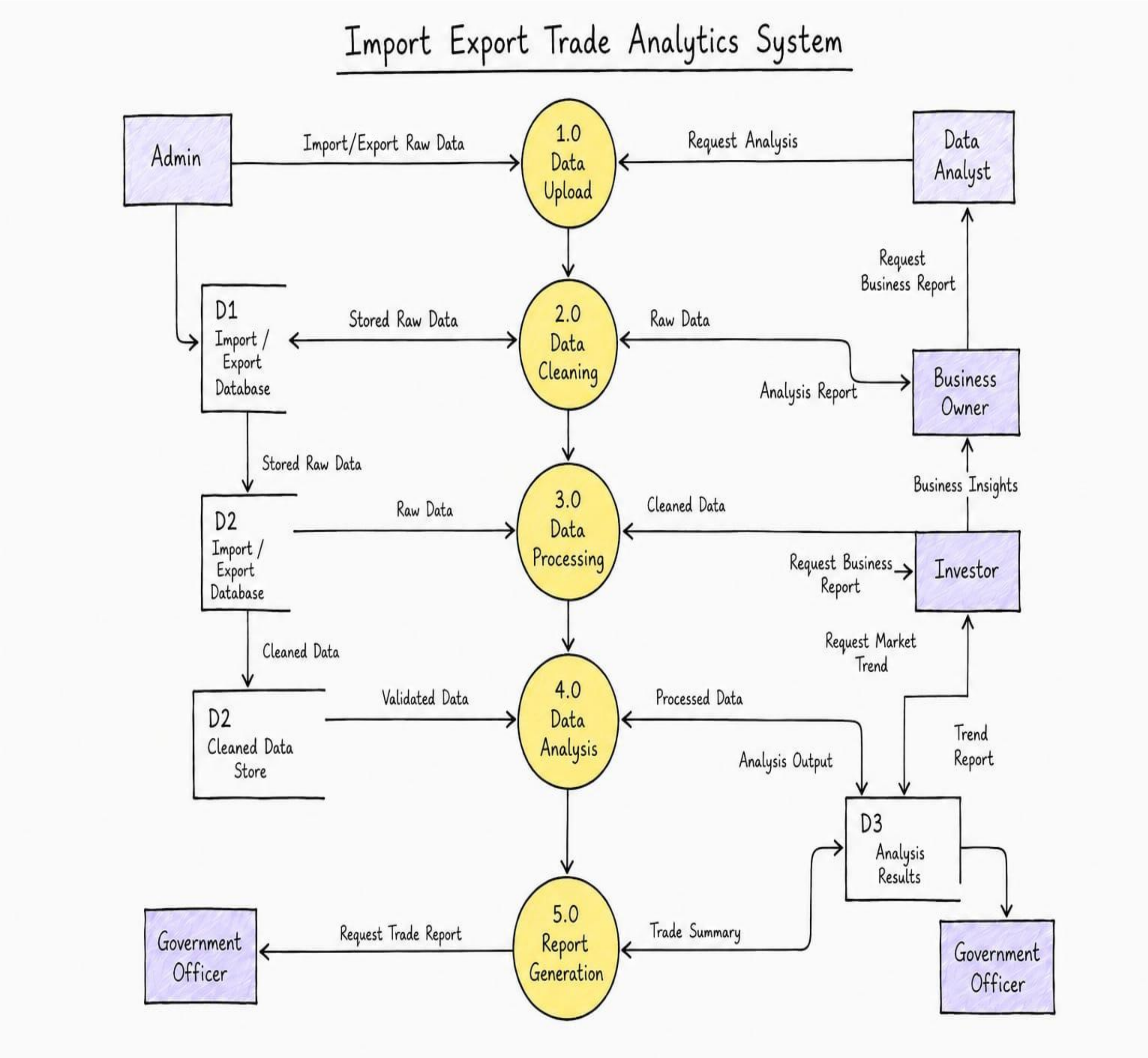
Data Flow:

Data flows from the TradeStat website to the data processing modules, where it is cleaned and analyzed. The processed data is stored in the database and displayed through charts, graphs, and dashboards for users and administrators.

DFD DESIGN: LEVEL 0



DFD DESIGN: LEVEL 1



6.2 ENTITY RELATIONSHIP (ER) DIAGRAM:

An Entity Relationship (ER) Diagram is a graphical representation of the database structure of a system. It shows the entities, attributes, and relationships between different data tables. ER diagrams are used during database design to provide a clear understanding of how data is stored, connected, and managed within the system.

In the Import Export Trade Analytics System, the ER diagram represents the relationship between products, HS codes, trade types, years, and trade records. It helps in organizing trade data efficiently for analytics, reporting, and decision-making purposes.

The ER diagram of the system contains the following major entities:

DIM_PRODUCT

DIM_HS2

DIM_TRADETYPE

DIM_YEAR

FACT_TRADE

These entities are connected through relationships such as classified, includes, and refers to.

1. SYMBOLS AND NOTATIONS USED IN ER DIAGRAM

The following symbols and notations are used in the ER Diagram of the Import Export Trade Analytics System:

Symbol	Notation Name	Description
▭	Entity	Represents a database table or entity
◯	Attribute	Represents fields or properties of an entity
◇	Relationship	Represents relationship between entities
<u>Attribute</u>	Primary Key (PK)	Unique identifier of a table
(FK)	Foreign Key	Attribute used to connect entities
1	One Cardinality	Represents one record
N	Many Cardinality	Represents multiple records
—	Connector Line	Connects entities and relationships
1 : N	One-to-Many Relationship	One entity connected to many records
M : N	Many-to-Many Relationship	Multiple records connected to multiple records
1 : 1	One-to-One Relationship	One record connected to one record

2.DESCRIPTION OF ENTITIES:

1. DIM_PRODUCT

The DIM_PRODUCT entity stores information related to products involved in import and export trade.

Attributes:

Product_ID (PK)

Commodity_Name

HS_Code

Industry_Category

Product_Type

Created_At

HS2_Code (FK)

This entity helps classify products according to industry categories and HS codes.

2. DIM_HS2

The DIM_HS2 entity contains HS2 classification details used for international trade categorization.

Attributes:

HS2_Code (PK)

Chapter_Name

Section_Name

Created_At

This entity helps maintain standardized product classification.

3. DIM_TRADETYPE

The DIM_TRADETYPE entity stores details about trade categories.

Attributes:

Trade_Type_ID (PK)

Trade_Type

Description

This table identifies whether the trade transaction belongs to import or export operations.

4. DIM_YEAR

The DIM_YEAR entity stores year and fiscal year details used for yearly trade analysis.

Attributes:

Year_ID (PK)

Fiscal_Year

Start_Date

End_Date

This entity helps perform year-wise trade comparisons and trend analysis.

5. FACT_TRADE

The FACT_TRADE entity is the central fact table containing trade analytics information.

Attributes:

Trade_ID (PK)

Product_ID (FK)

Year_ID (FK)

Trade_Type (FK)

Trade_Value

Share_Percentage

Growth_Rate

Absolute_Change

Trade_Balance

Created_At

This table stores measurable trade values and analytical information used in reports and dashboards.

3. RELATIONSHIPS IN ER DIAGRAM

1. Relationship between DIM_PRODUCT and DIM_HS2

A product belongs to one HS2 category, while one HS2 category can contain many products.

Relationship Type: One-to-Many (1:N)

2. Relationship between DIM_PRODUCT and FACT_TRADE

One product can appear in multiple trade records, while each trade record belongs to one product.

Relationship Type: One-to-Many (1:N)

3. Relationship between DIM_TRADETYPE and FACT_TRADE

One trade type can be associated with many trade records, while each trade record belongs to one trade type.

Relationship Type: One-to-Many (1:N)

4. Relationship between DIM_YEAR and FACT_TRADE

One year can contain multiple trade records, while each trade record belongs to one year.

Relationship Type: One-to-Many (1:N)

6.6 Advantages of ER Diagram in the System

The ER diagram provides several advantages in the Import Export Trade Analytics System:

Helps design a well-structured database

Reduces data redundancy

Maintains data consistency and integrity

Simplifies relationship management between entities

Supports analytics and reporting operations

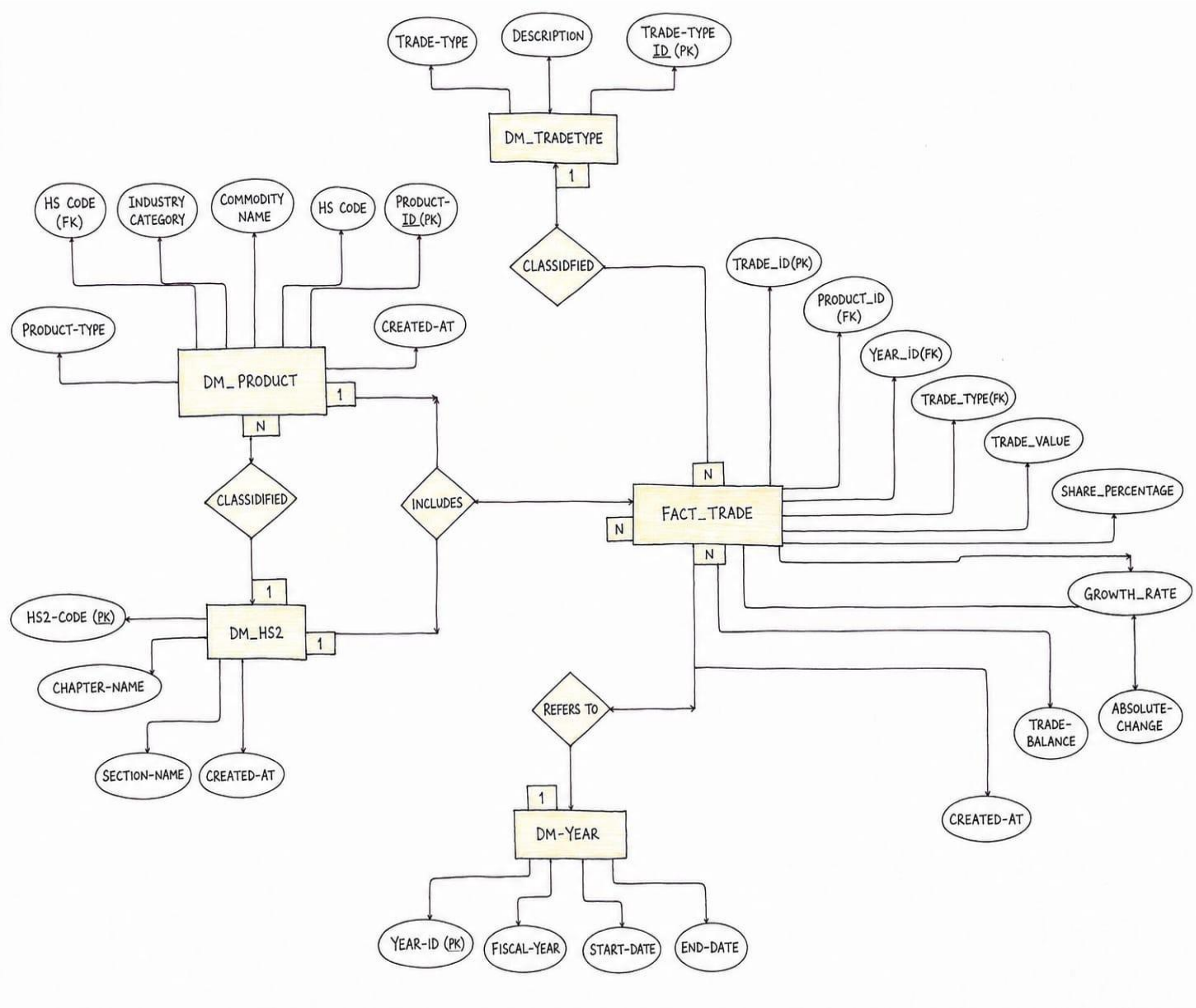
Improves database understanding and maintenance

Helps developers during implementation and testing

4. CONCLUSION

The ER Diagram of the Import Export Trade Analytics System provides a clear representation of the database structure and relationships between different entities. It plays an important role in designing an efficient database system for storing and analyzing import-export trade data. The use of entities, attributes, relationships, primary keys, and foreign keys ensures proper organization, consistency, and management of trade information within the system.

ER DAIGRAM:


TRADE-TYPE
DESCRIPTION
TRADE-TYPE ID (PK)
DM_TRADETYPE
1
CLASSIDFIED
HS CODE (FK)
INDUSTRY CATEGORY
COMMODITY NAME
HS CODE
PRODUCT-ID (PK)
PRODUCT-TYPE
CREATED-AT
DM_PRODUCT
1
N
CLASSIDIFIED
INCLUDES
1
DM_HS2
1
HS2-CODE (PK)
CHAPTER-NAME
SECTION-NAME
CREATED-AT
TRADE_ID(PK)
PRODUCT_ID (FK)
YEAR_ID(FK)
TRADE_TYPE(FK)
TRADE_VALUE
SHARE_PERCENTAGE
N
FACT_TRADE
N
N
GROWTH_RATE
TRADE-BALANCE
ABSOLUTE-CHANGE
CREATED-AT
REFERS TO
1
DM-YEAR
YEAR-ID (PK)
FISCAL-YEAR
START-DATE
END-DATE

6.3 TABLE DESIGN

Definition

Table design is the process of organizing and structuring data into tables so that information can be stored, managed, and retrieved efficiently. It forms the foundation of the database system and helps maintain data accuracy, consistency, and reliability.

In the Import Export Trade Analytics System, table design is used to store product details, HS code classifications, trade types, yearly trade records, trade values, growth rates, and analytics reports systematically.

Key Components:

Tables, Rows, and Columns

Tables store related information such as Product, HS2 Classification, Trade Type, Year, and Trade Analytics.

Rows represent individual records inside a table.

Columns define specific attributes such as product_id, hs2_code, trade_value, growth_rate, and trade_balance.

Data Types and Column Definition:

Each column is assigned a specific data type such as Integer, Varchar, Decimal, Date, or DateTime to ensure proper storage and consistency of data.

Primary Keys and Unique Constraints

Primary Key (PK) uniquely identifies each record in a table.

Unique Constraints prevent duplicate values and maintain data integrity.

Examples:

product_id

trade_id

year_id

hs2_code

Foreign Keys and Relationships:

Foreign keys are used to connect tables and establish relationships between entities.

Examples:

hs2_code links DIM_PRODUCT and DIM_HS2.

product_id links DIM_PRODUCT and FACT_TRADE.

year_id links DIM_YEAR and FACT_TRADE.

trade_type_id links DIM_TRADETYPE and FACT_TRADE.

Normalization:

Normalization organizes data into related tables to reduce redundancy and improve database efficiency and consistency.

Indexing:

Indexes improve the speed of searching and retrieving records from the database, helping the system process analytics, reports, and trade insights faster.

FACT_TRADE — Trade Fact Table

Column Name	Data Type	Key	Description
TRADE_ID	INT	PK	Unique trade record identifier
PRODUCT_ID	INT	FK	FK to DM_PRODUCT
YEAR_ID	INT	FK	FK to DM_YEAR
TRADE_TYPE	INT	FK	FK to DM_TRADETYPE
TRADE_VALUE	DECIMAL(18,2)		Total value of the trade
SHARE_PERCENTAGE	DECIMAL(5,2)		Share percentage of total trade
GROWTH_RATE	DECIMAL(5,2)		Year-over-year growth rate (%)
TRADE_BALANCE	DECIMAL(18,2)		Net trade balance (export minus import)
ABSOLUTE_CHANGE	DECIMAL(18,2)		Absolute change in trade value
CREATED_AT	DATETIME		Record creation timestamp

DM_TRADETYPE — Trade Type Dimension

Column Name	Data Type	Key	Description
TRADE_TYPE_ID	INT	PK	Unique trade type identifier
TRADE_TYPE	VARCHAR(50)		Name of trade type (e.g. Export, Import)
DESCRIPTION	VARCHAR(255)		Description of the trade type

DM_PRODUCT — Product Dimension

Column Name	Data Type	Key	Description
PRODUCT_ID	INT	PK	Unique product identifier
HS_CODE	VARCHAR(20)	FK	Harmonized System code (FK to DM_HS2)
COMMODITY_NAME	VARCHAR(255)		Name of the commodity/product
INDUSTRY_CATEGORY	VARCHAR(100)		Industry category of the product
PRODUCT_TYPE	VARCHAR(100)		Type/classification of the product
CREATED_AT	DATETIME		Record creation timestamp

DM_HS2 — HS2 Code Dimension

Column Name	Data Type	Key	Description
HS2_CODE	VARCHAR(10)	PK	2-digit Harmonized System chapter code
CHAPTER_NAME	VARCHAR(255)		Name of the HS2 chapter
SECTION_NAME	VARCHAR(255)		Name of the section under HS2
CREATED_AT	DATETIME		Record creation timestamp

DM_YEAR — Year Dimension

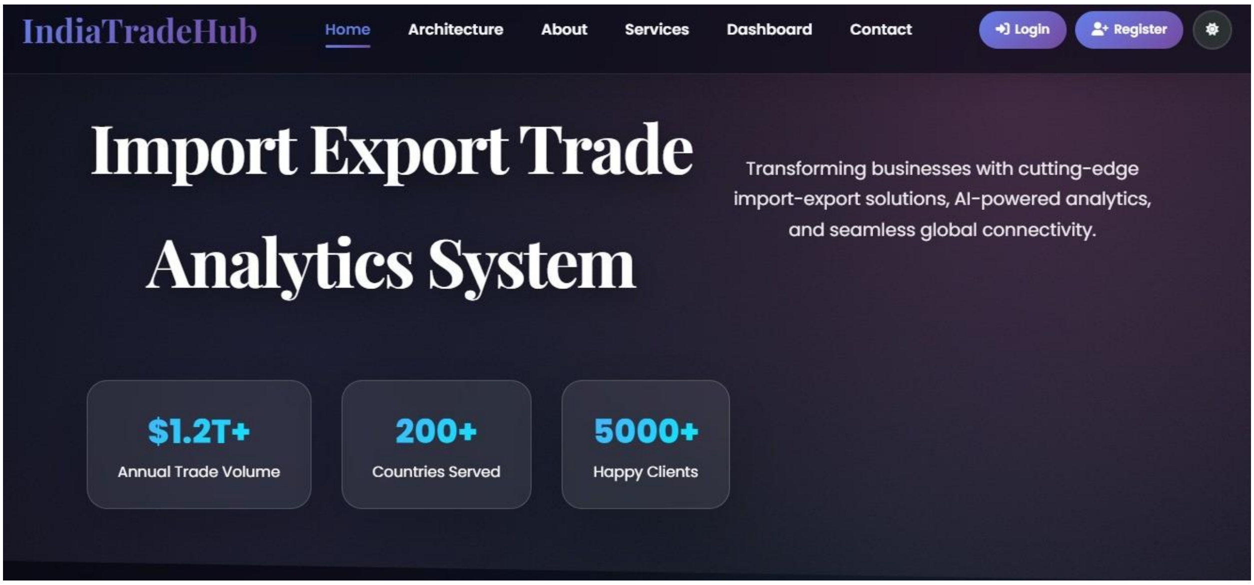
Column Name	Data Type	Key	Description
YEAR_ID	INT	PK	Unique year identifier
FISCAL_YEAR	VARCHAR(20)		Fiscal year label (e.g. 2023-24)
START_DATE	DATE		Start date of the fiscal year
END_DATE	DATE		End date of the fiscal year

Relationships

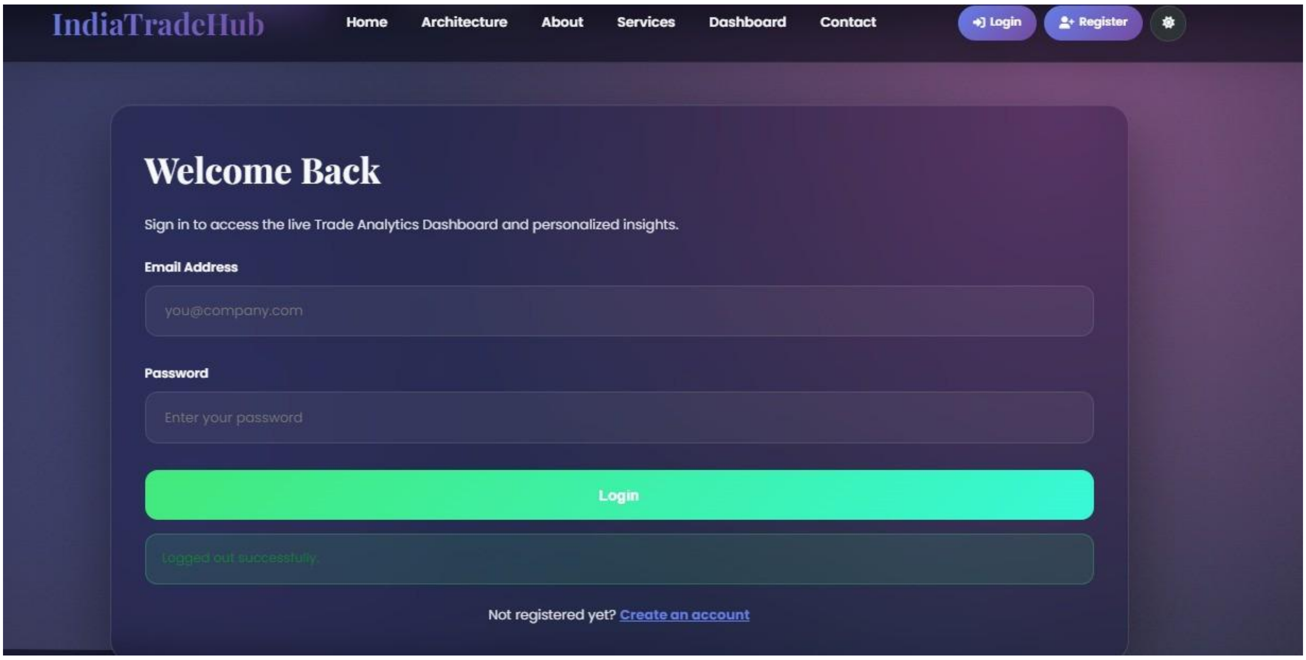
From Table	Cardinality	To Table	Via Column	Description
DM_TRADETYPE	1 → N	FACT_TRADE	TRADE_TYPE (FK)	One trade type → many trade records
DM_PRODUCT	1 → N	FACT_TRADE	PRODUCT_ID (FK)	One product → many trade records
DM_HS2	1 → N	DM_PRODUCT	HS_CODE (FK)	One HS2 chapter → many products
DM_YEAR	1 → N	FACT_TRADE	YEAR_ID (FK)	One year → many trade records

6.4FORM DESING

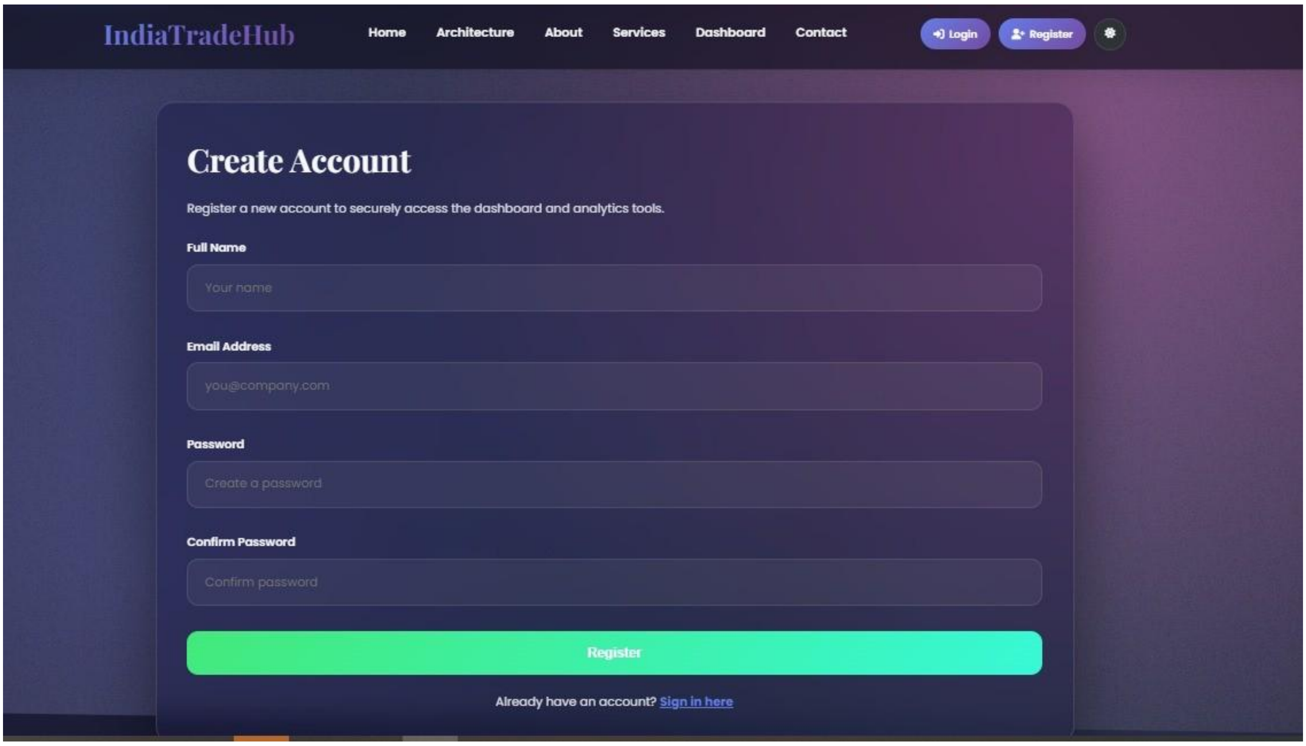
Home page



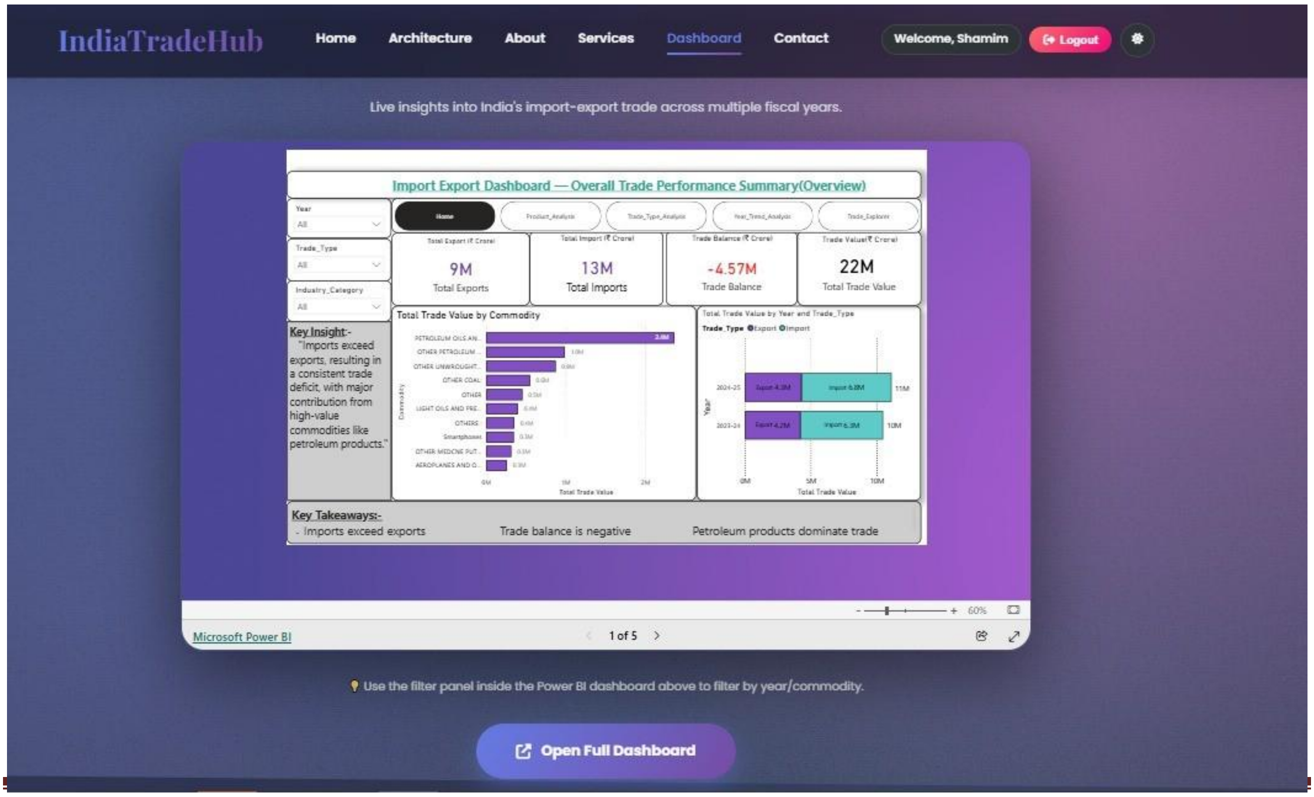
Login Page



Register Page



Dashboard



Coding

Python Script to Connect Mysql and Power Bi

```
Import pandas as pd
Import mysql.connector
# Connect to MySQL Database
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="Admin@123",
    database="import_export"
)
# Load all tables
fact_trade = pd.read_sql("SELECT * FROM fact_trade", conn)
dim_product = pd.read_sql(
    "SELECT * FROM dim_product", conn
)
dim_hs2 = pd.read_sql(
    "SELECT * FROM dim_hs2", conn
)
dim_year = pd.read_sql(
    "SELECT * FROM dim_year", conn
)
dim_tradetype = pd.read_sql(
    "SELECT * FROM dim_tradetype", conn
)
# Close connection
conn.close()
```

Home Page Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>IndiaTradeHub</title>
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.4.0/css/all.min.css">
<style>
:root{--pg:linear-gradient(135deg,#667eea,#764ba2);--ag:linear-
gradient(135deg,#4facfe,#00f2fe);-
-bg:#f5f7fa;--cb:#fff;--tp:#2d3436;--ts:#636e72;--sh:0 20px 40px rgba(0,0,0,.1)}
[data-theme=dark]{--bg:#0f0f23;--cb:rgba(45,52,54,.9);--tp:#f8f9fa;--ts:#dee2e6}
*{margin:0;padding:0;box-sizing:border-box}
body{font-family:Poppins,sans-serif;background:var(--bg);color:var(--tp);overflow-
x:hidden}
/* NAV */
.nav{position:fixed;top:0;width:100%;background:rgba(255,255,255,.55);backdrop-
filter:blur(20px);z-index:1000;padding:.8rem 0;border-bottom:1px solid
rgba(255,255,255,.18);box-
shadow:0 8px 35px rgba(0,0,0,.1)}
[data-theme=dark] .nav{background:rgba(10,10,25,.62)}
.nav-c{max-width:1400px;margin:0 auto;display:flex;justify-content:space-
between;align-
items:center;padding:0 2rem;gap:1rem}
.logo{font-size:1.9rem;font-weight:700;background:var(--pg);-webkit-background-
clip:text;-webkit-
text-fill-color:transparent;white-space:nowrap}
```

```
.links{display:flex;list-style:none;gap:2rem}
.links a{text-decoration:none;color:var(--tp);font-weight:600;font-
size:.95rem;transition:.3s}
.links a:hover{color:#667eea}
.nav-r{display:flex;align-items:center;gap:.5rem}
.btn{background:var(--pg);color:#fff;padding:.5rem 1rem;border-radius:50px;text-
decoration:none;font-weight:600;font-size:.85rem;border:none;cursor:pointer;white-
space:nowrap;transition:.3s}
.btn:hover{transform:translateY(-2px)}
.tgl{background:var(--cb);border:2px solid rgba(0,0,0,.08);border-
radius:50%;width:40px;height:40px;cursor:pointer;display:flex;align-
items:center;justify-
content:center;color:var(--tp);transition:.3s}
/* PROGRESS */
#prog{position:fixed;top:0;left:0;height:3px;width:0;background:linear-
gradient(90deg,#667eea,#f093fb);z-index:1001}
/* HERO */
.hero{min-height:100vh;padding-top:80px;background:linear-
gradient(135deg,rgba(0,0,0,.65),rgba(0,0,0,.35)),url('https://images.unsplash.com/pho
to-
1558618047-
3c8c76bbb17e?auto=format&fit=crop&w=2070&q=80')center/cover;display:flex;alig
n- items:center;justify-content:center;text-align:center;color:#fff}
.hero-c{max-width:900px;padding:0 1.5rem}
.hero-c h1{font-size:clamp(2.2rem,5vw,4.5rem);font-weight:900;margin-
bottom:1.2rem;text- shadow:0 10px 30px rgba(0,0,0,.5)}
.sub{font-size:1.1rem;margin-bottom:2rem;min-height:55px;opacity:.9}
.stats{display:flex;justify-content:center;gap:1.5rem;flex-wrap:wrap;margin-
bottom:2rem}
```

```
.stat{background:rgba(255,255,255,.1);backdrop-filter:blur(10px);padding:1.2rem
1.8rem;border- radius:20px;border:1px solid rgba(255,255,255,.2);transition:.3s}
.stat:hover{transform:scale(1.08);background:rgba(255,255,255,.2)}
.num{font-size:2rem;font-weight:800;background:var(--ag);-webkit-background-
clip:text;-webkit- text-fill-color:transparent}
.ctas{display:flex;gap:1rem;justify-content:center;flex-wrap:wrap}
.cp{background:var(--pg);color:#fff;padding:1rem 2.5rem;border-radius:50px;text-
decoration:none;font-weight:700}
.cs{color:#fff;padding:1rem 2.5rem;border-radius:50px;text-decoration:none;font-
weight:700;border:2px solid rgba(255,255,255,.5)}
.cp:hover,.cs:hover{transform:translateY(-3px)}
.cs:hover{background:rgba(255,255,255,.2)}
@media(max-width:768px){.links{display:none}.hero-c h1{font-size:2rem}}
</style>
</head>
<body>
<div id="prog"></div>
<nav class="nav">
<div class="nav-c">
<div class="logo">IndiaTradeHub</div>
<ul class="links">
<li><a href="#home">Home</a></li>
<li><a href="#architecture">Architecture</a></li>
<li><a href="#about">About</a></li>
<li><a href="#services">Services</a></li>
<li><a href="#dashboard">Dashboard</a></li>
<li><a href="#contact">Contact</a></li>
</ul>
<div class="nav-r">
```

```
<a href="#login" class="btn"><i class="fas fa-sign-in-alt"></i> Login</a>
<a href="#register" class="btn"><i class="fas fa-user-plus"></i> Register</a>
<button class="tgl" id="tgl"><i class="fas fa-moon"></i></button>
</div>
</div>
</nav>
<section class="hero" id="home">
<div class="hero-c">
<h1>Import Export Trade Analytics System</h1>
<p class="sub" id="sub"></p>
<div class="stats">
<div class="stat"><span class="num">$1.2T+</span><p>Annual Trade
Volume</p></div>
<div class="stat"><span class="num">200+</span><p>Countries Served</p></div>
<div class="stat"><span class="num">5000+</span><p>Happy Clients</p></div>
</div>
<div class="ctas">
<a href="#services" class="cp">Explore Services</a><a href="#dashboard"
class="cs">View Dashboard</a>
</div>
</div>
</section>
<script>
// Theme
const tgl=document.getElementById('tgl'); const
t=localStorage.getItem('theme')||'light'; document.body.setAttribute('data-theme',t);
tgl.querySelector('i').className=t==='dark'?'fas fa-sun':'fas fa-moon';
tgl.onclick=()=>{const n=document.body.getAttribute('data-
theme')==='dark'?'light':'dark';document.body.setAttribute('data-
```

```
theme',n);localStorage.setItem('theme',n);tgl.querySelector('i').className=n==='dark'
?'fas fa- sun':'fas fa-moon'};
// Scroll progress
window.onscroll=()=>document.getElementById('prog').style.width=(window.scrollY/
(document.bo dy.scrollHeight-innerHeight)*100)+'%';
// Typing
const txt="Transforming businesses with cutting-edge import-export solutions, AI-
powered analytics, and seamless global connectivity.";
const el=document.getElementById('sub');let i=0;
setTimeout(()=>{(function
t(){if(i<txt.length){el.textContent+=txt[i++];setTimeout(t,30)}})()},800);
</script>
</body>
</html>
```

Login page

```
<!-- ========== LOGIN ========== -->
<section id="login" class="auth-section">
<div class="auth-container">
<div class="auth-card" data-aos="fade-up">
<h2>Welcome Back</h2>
<p>Sign in to access the live Trade Analytics Dashboard and personalized
insights.</p>
<form id="loginForm">
<div class="form-group">
<label for="loginEmail">Email Address</label>
<input type="email" id="loginEmail" placeholder="you@company.com" required>
</div>
<div class="form-group">
<label for="loginPassword">Password</label>
```

```
<input type="password" id="loginPassword" placeholder="Enter your password" required minlength="6">
</div>
<button type="submit" class="submit-btn">Login</button>
<div id="loginMessage" class="alert-box" style="display:none;"></div>
</form>
<div class="auth-switch">
Not registered yet? <a href="#register" style="color:#667eea; font-weight:700;">Create an account</a>
</div>
</div>
</div>
</section>
```

Register page

```
<!-- ========== REGISTER ========== -->
<section id="register" class="auth-section">
<div class="auth-container">
<div class="auth-card" data-aos="fade-up">
<h2>Create Account</h2>
<p>Register a new account to securely access the dashboard and analytics tools.</p>
<form id="registerForm">
<div class="form-group">
<label for="registerName">Full Name</label>
<input type="text" id="registerName" placeholder="Your name" required>
</div>
<div class="form-group">
<label for="registerEmail">Email Address</label>
<input type="email" id="registerEmail" placeholder="you@company.com" required>
```

```
</div>
<div class="form-group">
<label for="registerPassword">Password</label>
<input type="password" id="registerPassword" placeholder="Create a password" required minlength="6">
</div>
<div class="form-group">
<label for="registerConfirm">Confirm Password</label>
<input type="password" id="registerConfirm" placeholder="Confirm password" required>
</div>
<button type="submit" class="submit-btn">Register</button>
<div id="registerMessage" class="alert-box" style="display:none;"></div>
</form>
<div class="auth-switch">
Already have an account? <a href="#login" style="color:#667eea; font-weight:700;">Sign
in here</a>
</div>
</div>
</div>
</section>
```

Dashboard page

```
</div><!-- ========== DASHBOARD ========== -->
<div class="dashboard-section-wrapper">
<section id="dashboard" class="dashboard-container">
<div class="dashboard-locked" id="dashboardLocked">
<i class="fas fa-lock lock-icon"></i>
<h3 style="font-family:'Playfair Display',serif; font-size:2rem;">Dashboard Access
```

```
Required</h3>
<p style="color:var(--text-secondary); max-width:500px; margin:1rem 0 2rem;">
Please sign in or register to view the live Trade Analytics Dashboard.
</p>
<div style="display:flex; gap:1rem; flex-wrap:wrap; justify-content:center;">
<a href="#login" class="login-btn" style="padding:1rem 2rem; font-size:1rem;">
<i class="fas fa-sign-in-alt"></i> Login
</a>
<a href="#register" class="login-btn" style="padding:1rem 2rem; font-size:1rem;">
<i class="fas fa-user-plus"></i> Register
</a>
</div>
</div>
<div class="dashboard-content" id="dashboardContent">
<h2 class="section-title" data-aos="zoom-in">Trade Analytics Dashboard</h2>
<p style="text-align:center; color: var(--text-secondary); margin-bottom: 1.5rem;">
Live insights into India's import-export trade across multiple fiscal years.
</p>
<div class="powerbi-wrapper" data-aos="zoom-in">
<div class="dashboard-loading" id="dashboardLoading">
<div class="loading-spinner"></div>
<p style="color:var(--text-secondary);">Loading dashboard...</p>
</div>
<iframe id="powerbiFrame"
src="https://app.powerbi.com/view?r=eyJrIjoiMGRhMWQ3NjctNTVkZi00NzhkLTl
jYWUtMzg5ZWI5Y
TZjNjRmIiwidCI6IjVmOTA5NjYwLWUxMDQtNDNkNC1hOGViLTA2YjNjNTkx
YjBiMSJ9"
class="powerbi-embed"
```

```
allowfullscreen
loading="lazy"
title="India TradeHub Dashboard"
onload="hideDashboardLoader()"></iframe>
</div>
<p style="font-size:0.9rem; color:var(--text-secondary); margin-bottom:2rem; text-align:center;">
</p>
□ Use the filter panel inside the Power BI dashboard above to filter by year/commodity.
<div style="display:flex; gap:1rem; justify-content:center; flex-wrap:wrap;">
<a
href="https://app.powerbi.com/view?r=eyJrIjoiMGRhMWQ3NjctNTVkZi00NzhkLT
ljYWUtMzg5ZWI5
YTZjNjRmIiwidCI6IjVmOTA5NjYwLWUxMDQtNDNkNC1hOGViLTA2YjNjNT
kxYjBiMSJ9"
target="_blank" class="dashboard-btn">
<i class="fas fa-external-link-alt"></i> Open Full Dashboard
</a>
</div>
<div class="dashboard-features">
<div class="dash-feature-tag"><i class="fas fa-circle" style="color:#43e97b; font-size:8px;"></i> 5 Pages</div>
<div class="dash-feature-tag"><i class="fas fa-filter"></i> Interactive Filters</div>
<div class="dash-feature-tag"><i class="fas fa-rupee-sign"></i> ₹ Crore</div>
<div class="dash-feature-tag"><i class="fas fa-calendar"></i> FY 2023-24 & 2024-25</div>
<div class="dash-feature-tag"><i class="fas fa-boxes"></i> 6,144 Commodities</div>
```

```
</div>
</div>
</section>
```

SYSTEM IMPLEMENTATION AND TESTING

The system implementation phase involves developing and integrating all the components required for the project. The *Import Export Trade Analytics System* is implemented using Python, MySQL, HTML5, CSS3, JavaScript, and Microsoft Power BI. The system is designed to collect, process, analyze, and visualize import–export trade data efficiently.

7.1 System Implementation

Import–export trade data is collected from the TradeStat website.

The collected data is cleaned and processed using Python libraries such as Pandas and NumPy.

MySQL is used to store and manage the processed data in a structured format.

HTML5, CSS3, and JavaScript are used to design a simple and user-friendly interface.

The system includes **user registration and login pages** for secure access.

Users must log in successfully to access the dashboard and system features.

Power BI is used to generate charts, graphs, and dashboards for better visualization of trade data.

The system helps users understand trade trends, country-wise analysis, and product performance effectively.

7.2 System Testing

The system is tested to ensure that all functions work properly and provide accurate results.

The registration and login pages are tested to verify secure user access.

The dashboard is checked to ensure it opens only after successful login.

The data collection process is tested to verify that trade data is collected correct

MERITS & DEMERITS

8.1 MERITS (ADVANTAGES):

Automated data analysis:

The system provides automated data analysis, which reduces the need for manual work and increases efficiency.

Faster and more accurate results:

The system processes data quickly and provides accurate results compared to manual methods.

Efficient handling of large datasets:

It can handle large volumes of trade data effectively without performance issues.

Improved data visualization:

The system uses charts and graphs to represent data clearly, making it easier to understand trends and patterns.

Reduced manual effort:

By automating tasks such as data collection and analysis, the system minimizes human effort and saves time.

8.2 DEMERITS (DISADVANTAGES):

Dependence on availability of data:

The system depends on the availability of data from the TradeStat website. If data is not available or updated, the analysis may be affected.

Requires stable internet connection:

A stable internet connection is necessary to collect and update data. Without internet access, the system cannot function properly.

FUTURE ENHANCEMENT

Future enhancement refers to the possible improvements that can be made to the system to increase its performance and usability.

Integration of real-time trade data:

The system can be enhanced by adding real-time data integration, which will provide updated and accurate trade information.

Use of machine learning for prediction:

Machine learning techniques can be used to predict future trade trends and patterns, helping in better decision-making.

Development of a web-based dashboard:

A web-based dashboard can be developed to make the system more accessible and user-friendly.

Expansion to multiple data sources:

The system can be extended to collect data from multiple sources, improving data availability and analysis.

CONCLUSION

The project demonstrates the use of data analytics to analyze import–export trade data collected from the TradeStat website. Data processing and visualization improve efficiency and accuracy, making the system useful for understanding global trade patterns.

This project helps in overcoming the limitations of manual data analysis by providing an automated solution. It reduces the time and effort required to collect and process large volumes of trade data. The system also improves accuracy by minimizing human errors.

The use of graphical visualization techniques such as charts and graphs makes it easier to understand complex data. It helps users to identify trends, patterns, and relationships in trade activities more effectively.

Overall, the system is useful for analyzing trade data and supports better decision-making. It provides a simple and efficient way to understand import and export patterns.

In conclusion, the project successfully achieves its objectives and can be further improved with additional features in the future.

BIBILOGRAPHY

Technology	Author Name	Text Book / Reference	Edition
Python	Wes McKinney	*Python for Data Analysis*	2nd Edition Chapter 3,5 Page no 44-120
MySQL	Oracle Corporation	*MySQL Reference Manual*	Official Documentation Chapter 2 Page no 15-60
Power BI	Microsoft Corporation	*Microsoft Power BI Documentation*	Official Documentation Chapter 4 Page no 80-140
Pandas	Pandas Development Team	*Pandas Documentation*	Official Documentation Chapter 1,2 Page no 10-55

Official Documentation

Python Official Documentation –https://docs.python.org/3/

Pandas Documentation –https://pandas.pydata.org/docs/

MySQL Documentation –https://dev.mysql.com/doc/

Microsoft Power BI Documentation –https://learn.microsoft.com/power-bi/

TradeStat Official Website –https://tradestat.commerce.gov.in/

YouTube Channels

CodeBasics –https://www.youtube.com/@codebasics

freeCodeCamp.org –https://www.youtube.com/@freecodecamp

AI Assistant

ChatGPT by OpenAI –https://chat.openai.com